University of Ottawa

Department of Mathematics and Statistics

MAT 1302C : Mathematical Methods II

Professor: Hadi Salmasian

Second Midterm Exam – Version A

November 22, 2011

Surname _____ First Name _____

Student # _____ DGD (Tuesday/Thursday) _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) The number of points available for each question is indicated in square brackets.
(c) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
(d) Write your student number at the top of each page in the space provided.
(e) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(f) You are strongly recommended to write in **pen**, not pencil.
(g) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	4	5	4	4	3	4	24
Grade							

1. (a) **(3 points)** Use cofactor expansion across a row to compute the determinant of the following matrix. (You can choose any row that you want.)

$$A = \begin{bmatrix} 1 & 0 & -2 & 1 \\ 2 & 0 & 0 & -1 \\ 1 & 3 & 5 & 1 \\ 3 & -1 & 1 & 2 \end{bmatrix}$$

(b)**(1 point)** Is A invertible? Justify your answer.

2. (a) **(3 points)** Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 4 & -\frac{1}{2} & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 0 & 0 \\ 0 & -\frac{1}{2} & -\frac{4}{5} \\ 0 & 0 & -1 \end{bmatrix}$. Suppose that a 3×3 matrix C satisfies

$$B^{-1}A(C^T)^{-1}B^2 = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Compute $\det C$. Justify your answer.

(b) **(2 points)** Suppose a, b, c are three numbers such that $\begin{vmatrix} a & 1 & 3 & 1 \\ b & 0 & -2 & \frac{1}{3} \\ -2 & 3 & 5 & 1 \\ c & \frac{1}{2} & -5 & 2 \end{vmatrix} = 5$. What is the value of $\begin{vmatrix} a+2 & 1 & 3 & 3 \\ b & 0 & -2 & 1 \\ 4 & 3 & 5 & 3 \\ c+1 & \frac{1}{2} & -5 & 6 \end{vmatrix}$? Justify your answer.

3. Consider the matrix

$$A = \begin{bmatrix} 3 & -6 & 0 & 4 & 0 \\ 1 & -2 & 1 & 5 & -1 \\ 1 & -2 & 0 & 2 & 0 \end{bmatrix}.$$

(a) **(3 points)** Find a basis for $\operatorname{Nul} A$.

(b) **(1 point)** Determine the rank of A. Justify your answer.

4. **(4 points)** Determine if the following matrix is invertible. If it is invertible, then find its inverse.

$$\begin{bmatrix} 1 & -2 & 0 \\ 1 & -1 & -3 \\ 1 & -2 & 1 \end{bmatrix}$$

5. Determine if the following subsets of $\mathbb{R}^3$ are subspaces or not. Justify your answers.

(a) **(1 point)** $H = \left\{ \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \;\middle|\; x, y \text{ are in } \mathbb{R} \right\}$.

(b) **(1 point)** $H = \left\{ \begin{bmatrix} x+2y \\ -3y \\ x-2y \end{bmatrix} \;\middle|\; x, y \text{ are in } \mathbb{R} \right\}$.

(c) **(1 point)** $H = \left\{ \begin{bmatrix} x \\ y \\ z^2 \end{bmatrix} \;\middle|\; x, y, z \text{ are in } \mathbb{R} \right\}$.

6. An economy has two sectors: Manufacturing and Service. In order to produce one unit of output, Manufacturing requires .3 units from its own sector and .6 units from Service. On the other hand, Service requires .4 units from its own sector and .2 units from Manufacturing to produce one unit of output.

(a) **(1 point)** Write down the consumption matrix C for this economy.

(b) **(1 point)** Determine the intermediate demands if Service decides to produce 15 units and Manufacturing decides to produce 20 units.

(c) **(2 points)** Determine the production levels required to meet a final demand of 15 units from Manufacturing and 9 units from Service.

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